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Stress in remote work: two studies testing the Demand-Control-Person model

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ABSTRACT

The popularity of remote work continues to rise, but uncertainty remains about how it influences employee well-being. We extend the Demand-Control-Person (DCP) model to test both person and job factors as important considerations in remote work, suggesting that emotional stability influences the utility of autonomy as a job resource in protecting employees from strain. Additionally, we test self-determination theory (SDT), positioning need satisfaction for autonomy, relatedness, and competence as mechanisms explaining the relationship between remote work and strain. In two field studies, high–emotional stability employees reporting high levels of autonomy experienced the lowest levels of strain, with negative relationships between extent of remote work and strain. In contrast, low–emotional stability employees who also have high autonomy appear more susceptible to strain, and this may increase when they work remotely more often. Our multilevel structural equation modelling revealed that high–emotional stability employees with high autonomy appear best positioned to meet their needs for autonomy and relatedness, even when remote work is more frequent; these in turn reduced the likelihood of strain. Thus, our results support the DCP and SDT models, revealing theoretical and practical implications for designing and managing remote work arrangements.

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Strain; stress; remote work; Demand-Control-Person model; self-determination theory

The workplace is changing in response to a number of technological and societal trends that are making work more accessible and the execution of work faster, cheaper, and greener. As predicted more than three decades ago by Shamir and Salomon (1985), organizations now offer employees more flexibility than ever, including in the form of remote work arrangements (GlobalWorkplaceAnalytics.com, 2018; Thompson, Payne, & Taylor, 2015). Remote work (i.e., performing work at a location other than one's primary office) is one form of flexibility that is highly valued by many and often used by organizations to attract high-quality applicants (Allen, Johnson, Kiburz, & Shockley, 2012; Humphrey, Nahrgang, & Morgeson, 2007; Thompson et al., 2015). Recent estimates indicate that 20–25% of employees worked remotely to some extent in 2015, and these numbers grew by 115% between 2005 and 2015 (GlobalWorkplaceAnalytics.com, 2018). In their most recent study, the Society for Human Resource Management (SHRM) found that 59% of 2,700 firms across industries offer employees the opportunity to work remotely on an ad-hoc basis, up from 54% in 2014 (Allen, Golden, & Shockley, 2015; SHRM, 2017). Furthermore, 23% of firms offer full-time remote work arrangements, and 35% offer part-time arrangements (SHRM, 2017).

This shift suggests that people and organizations find value in remote work, begging the question: how does the extent to which one engages in remote work impact well-being? The overwhelming conclusion from over a decade of empirical research on the linkage between remote work and employee well-being is that "it depends" (e.g., Allen et al., 2012; Gajendran & Harrison, 2007; Giberson & Miklos, 2013; Michel, Kotrba, Mitchelson, Clark, & Baltes, 2011). Remote work may function as a challenge stressor, which is both energy-depleting and stimulating; indeed, it is positively linked to both burnout and engagement (Crawford, LePine, & Rich, 2010; van den Broeck, de Cuyper, de Witte, & Vansteenkiste, 2010). Because working remotely impacts the physical, social, and psychological context of work, more extensive remote work can be depleting due to the challenges it presents and resources it consumes (e.g., time spent coordinating with others, managing ambiguity or managing non-work stressors, loss of socio-emotional support), causing increased strain (Crawford et al., 2010; Giberson & Miklos, 2013; Sewell & Taskin, 2015). Despite the challenges, however, remote work also stimulates employees by offering important benefits (Giberson & Miklos, 2013). For example, studies suggest that employees with location flexibility experience increased satisfaction and commitment and reduced turnover (Golden, 2006a; Hunton & Norman, 2010), lower work–family conflict (Rousseau, Ho, & Greenberg, 2006), a desire to work harder (Gajendran, Harrison, & Delaney-Klinger, 2015; Kelliher & Anderson, 2010), and are better able to manage the demands of long working hours (Duxbury & Halinski, 2014).

As is evident in this brief snapshot of the literature, Shamir and Salomon (1985) were right in their early predictions that not all remote work arrangements would be equal, depending on the type of job and employee involved. They compared the jobs of closely supervised clerical employees performing routine tasks to information analysts autonomously performing complex knowledge work, noting that differences in autonomy between these two types of jobs are likely exacerbated when they are

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performed remotely. Empirical research on the importance of autonomy and other job and person characteristics in remote work is emerging to support this notion (Barsness, Diekmann, & Seidel, 2005; Gajendran & Harrison, 2007; Sardeshmukh, Sharma, & Golden, 2012). We aim to contribute to this line of work by untangling the independent and interactive effects of the extent to which an employee works physically away from the primary office (ranging from none to full-time) and the job autonomy (i.e., discretion) available to employees. Applying the Demand-Control-Person (DCP) model (Rubino, Perry, Milam, Spitzmüller, & Zapf, 2012), an extension of the classic Demand-Control (D-C) model (Karasek, 1979), and extending research that positions emotional stability as a "key" resource (Halbesleben, Neveu, Paustian-Underdahl, & Westman, 2014; Meier, Semmer, Elfering, & Jacobshagen, 2008; Ten Brummelhuis & Bakker, 2012), we also suggest that emotional stability affects the way autonomy is used as a resource, jointly determining the strain outcomes of the extent of remote work.

In developing and testing three moderated mediation models and performing hypothesis tests in two independent field studies, we aim to contribute to theory and practice regarding remote work, personality, and employee well-being. This represents an early effort to test and extend the DCP model (Pisanti et al., 2015; Rubino et al., 2012) by examining the joint role of emotional stability and autonomy in the context of remote work. We also contribute to a burgeoning line of research seeking to understand the role of autonomy in conjunction with other resources in this context, namely, emotional stability (e.g., Gajendran et al., 2015; Golden, 2012; Lapierre, Van Steenbergen, Peeters, & Kluwer, 2016). This addresses Shamir and Salomon's (1985) early concerns about remote work and employee well-being across varying levels of autonomy. Our emphasis on the interactive role of the person and job factors is another contribution that directly addresses Allen et al.'s (2015) suggestion for a multifaceted approach to the study of remote work. Thus, our aim is to highlight the unique ways in which different employees may react to varying levels of remote work and the level of autonomy offered.

Finally, we also contribute to the literature by taking a more theoretical approach than has typically been used in the study of remote work (Gajendran & Harrison, 2007). By applying the D-C (Karasek, 1979) and DCP (Rubino et al., 2012) models, we carefully explicate specific boundary conditions of the extent of remote work–strain relationship. Further, applying self-determination theory (SDT; Deci & Ryan, 2000), we examine three mechanisms by which job and person factors may influence employee well-being as the extent of remote work increases—employees' perceived need satisfaction in regard to autonomy, relatedness with co-workers, and competence.

The D-C model in remote work

Before reviewing the literature, it is necessary to first be clear about how we conceptualize remote work and delineate it from other similar constructs. Aligning our work with Allen et al. (2015), we define the extent of remote work as the frequency with which an employee works away from their primary office at any other location, ranging from none (i.e., traditional office-based work) to full-time (Bailey & Kurland, 2002; Gajendran & Harrison, 2007). Adopting remote work as a broader label than telecommuting or telework, we include any work location that is not a primary office operated by the organization, such as home, coffee shops, or customer sites (Allen et al., 2015; Gajendran & Harrison, 2007). Although remote work locations could also, in theory, include satellite offices, these may still be considered a "primary" office location according to our conceptualization, since the conveniences and social context of office resources and other co-workers are likely present in those locations (Allen et al., 2015; Thatcher & Zhu, 2006). Also, we include traditional office-based workers and part-time to full-time remote workers in our sample, because this range captures the reality of most workplaces.

Moreover, in contrast to research on idiosyncratic deals (i-deals) or flexible work arrangements, we consider one's autonomy in typical aspects of a job (i.e., process, methods) in conjunction with only one form of flexible work arrangement (i.e., location), rather than including other aspects of work that can be customized for individual employees (e.g., flexible schedules or training programmes; Allen et al., 2015; Rousseau et al., 2006; Rousseau, Tomprou, & Simosi, 2016). Of course, i-deals can include negotiated remote work arrangements, among other job design factors, but i-deals are not the focus of our study.

Strain in remote work

Strain, a diminished state of well-being, is the outcome of interest in this study. The D-C model proposes that job demands and control interact to predict strain, wherein individuals who experience high demands with little personal control will experience the most negative health consequences (Karasek, 1979). Furthermore, strain occurs when resources (i.e., energy, support, time, personal characteristics) are threatened or depleted, thus preventing individuals from effectively managing demands, achieving work goals, or engaging in personal and professional development (Crawford et al., 2010; Halbesleben & Demerouti, 2005; Halbesleben et al., 2014; Hobfoll, 2001).

The extent to which an employee works remotely is likely to have an influence on strain because it describes how often an employee is in a work environment that is different from the typical office environment where co-workers, leaders, and physical resources are present. Although greater flexibility in general has been linked to higher levels of commitment, satisfaction, and performance (Gajendran et al., 2015; Golden, 2006a; Kelliher & Anderson, 2010), the extent of remote work is likely to function as a challenge stressor, positively impacting job attitudes, performance, and motivation but also potentially inducing employee strain (van den Broeck et al., 2010). Despite the stimulating effect greater flexibility provides, differences between primary and remote work locations can be taxing, as well as impact the level of resources available and their ability to use them (Sardeshmukh et al., 2012). Indeed, it is likely that those who work remotely more often experience more strain (Giberson & Miklos, 2013) when they have to expend more time and energy contacting colleagues for information or approvals, coordinating on tasks, and even completing basic tasks without access to office technology and equipment (Rousseau et al., 2016). They may also miss out on resources shared in spontaneous employee interactions in the office, such as socio-emotional support, information, or the visibility that comes from simply being in the office (Dennis & Wixom, 2002). Less exposure to co-workers and leaders may also present challenges, as one's relative level of status may be unclear (Bailey & Kurland, 2002), and expectations may be more ambiguous (Sardeshmukh et al., 2012). Challenges associated with working at home in particular further include the need to balance both home and work roles (McNaughton, Rackensperger, Dorn, & Wilson, 2014; Shumate & Fulk, 2004), to manage non-work-related stressors even while "at work" (Konradt, Hertel, & Schmook, 2003; Sardeshmukh et al., 2012), and a lack of recovery time from both work and non-work stressors (Golden, 2012). Workers who are often remote may also have more non-traditional work hours, and they may experience job intensification as a result of fewer physical or psychological boundaries between home and work (Golden, 2012; Kelliher & Anderson, 2010). As such, we position remote work as a challenge demand that could result in strain, particularly when the appropriate resources are not available.

In this study, we consider three different forms of strain as outcomes, which are all key components of affective well-being (Mäkikangas, Kinnunen, Feldt, & Schaufeli, 2016): exhaustion, disengagement, and dissatisfaction. Though dissatisfaction and exhaustion are the most often used in testing the D-C model in the existing literature (e.g., Luchman & González-Morales, 2013), recent studies of the D-C (e.g., Kühnel, Sonnentag, & Bledow, 2012; Mauno, Mäkikangas, & Kinnunen, 2016) and the DCP (e.g., Peters et al., 2016; Rubino et al., 2012) models also include disengagement. Thus, we are able to directly build on the current literature and extend that literature by using these three outcomes.

Exhaustion describes the overall level of psychological and affective energy for engaging in work-related tasks (Halbesleben & Buckley, 2004). Disengagement is a complementary but distinct affective-motivational construct that describes the level of energized, dedicated involvement in one's work. Engaged individuals fully invest themselves and their resources in their work and the workplace, whereas disengaged employees are detached, withdrawn, and avoid significant investments in work (Christian, Garza, & Slaughter, 2011). Finally, dissatisfaction is a negative evaluative attitude developed as employees make judgements about their overall work experience (Cammann, Fichman, Jenkins, & Klesh, 1983).

Of these three forms of strain, exhaustion is most closely related to physical and psychological health, but may also have a motivational component like disengagement and dissatisfaction. All three are related to performance and turnover (Christian et al., 2011; Wright & Cropanzano, 1998) and are thus important considerations. Research suggests that a negative relationship exists between the extent of remote work and both exhaustion and disengagement (Gajendran & Harrison, 2007; Sardeshmukh et al., 2012), and that autonomy and other resources are mediators of this relationship. In contrast, moderate levels of remote work are thought to reduce dissatisfaction, but this also depends on other moderating factors, including autonomy (Allen et al., 2015; Golden & Viega, 2005). This suggests there may be nuances in how the extent of remote work associates with each strain outcome, making each worthy of consideration. The distinctions between these three constructs may be more salient when considering their outcomes; this is another reason to consider each individually, even though we expect directionally similar effects of our proposed predictors on each form of strain. For the sake of brevity, we refer to strain in general as we develop our hypotheses later.

Autonomy as a resource in remote work

Autonomy is considered a valuable resource across jobs, but scholars have identified it as particularly critical for success in remote work, including for fostering employee well-being (Gajendran & Harrison, 2007; Lapierre et al., 2016; Sardeshmukh et al., 2012). Indeed, meta-analytic findings suggest that autonomy is "one of the principal mechanisms through which telecommuting had its positive attitudinal and behavioral effects" (Gajendran & Harrison, 2007, p. 1534). Proponents of remote work advocate for the benefits that increased autonomy provides, yet autonomy is not an inherent part of remote work (Dimitrova, 2003). Although workers may perceive they have greater autonomy given their flexibility in choosing *where* they work, they may still have limited autonomy regarding *how* they accomplish tasks. The level of autonomy can vary based on job level (e.g., manager versus team member) and across employers, as some employers provide more specific and detailed task requirements, performance standards, and electronic monitoring processes when employees work remotely compared to on-site workers (Ambrose & Adler, 2000). Managers overseeing remote workers may even use a more controlling management style, which implies less trust in employees (Boell, Cecez-Kecmanovic, & Campbell, 2016; Clear & Dickson, 2005; Hunton, 2005; Sewell & Taskin, 2015). Still, the ability to work remotely is often valued by employees, as it offers one form of autonomy (i.e., location; Gajendran & Harrison, 2007; Harmer & Pauleen, 2012).

The D-C model (Karasek, 1979) positions autonomy as one aspect of job control (Karasek & Theorell, 1990), suggesting that it may help employees cope with the challenges of working remotely and appropriately leverage its advantages (Gajendran et al., 2015; Halbesleben et al., 2014; Karasek & Theorell, 1990). While Gajendran and Harrison's (2007) meta-analysis found that remote work increases perceptions of autonomy, it did not examine how autonomy affects the relationship between remote work and well-being outcomes. Based on the D-C model, employees with higher levels of autonomy are empowered to productively address the challenges involved in remote work and deploy other resources available to them in ways that suit the situation, thereby protecting their well-being (Bailey & Kurland, 2002; Chung-Yan, 2010; de Jonge & Kompier, 1997; Sardeshmukh et al., 2012). In contrast, lower autonomy may exacerbate the challenges of working remotely, reducing an employee's options for managing such challenges. Thus, in line with the classic D-C model, we first formally propose that autonomy will moderate the relationship between extent of remote work and all three forms of strain, as follows:

Hypothesis 1. Autonomy is a negative moderator of the extent of remote work–strain relationship, such that lower levels of autonomy result in a positive relationship, and higher levels of autonomy result in a negative relationship.

Caveat to the D-C model: personality in remote work

In addition to autonomy, the DCP model is consistent with other resource theories in suggesting that personality determines how individuals perceive, react to, manage, and experience resources, stress, and strain (e.g., Crawford et al., 2010; Halbesleben et al., 2014; Hobfoll, 2001; Quinn, Spreitzer, & Lam, 2012). Specifically, building on research that shows the D-C model only holds for certain personality traits (e.g., proactivity, emotional stability, internal locus of control; de Jonge & Kompier, 1997; Meier et al., 2008; Parker & Sprigg, 1999), the DCP model incorporates emotional stability as an additional moderator, positing that this trait may equip individuals to deploy available autonomy resources (Costa & McCrae, 1988; Rubino et al., 2012). We extend this to suggest individuals with higher levels of emotional stability are better equipped to manage both the challenges and advantages inherent in remote work.[1]

Individuals high in emotional stability generally waste fewer resources than those with low emotional stability because they are more calm, confident, and optimistic (Costa & McCrae, 1988; Kammeyer-Mueller, Simon, & Judge, 2016), thereby experiencing the lowest levels of exhaustion and other forms of strain (Halbesleben & Buckley, 2004). Thus, in line with the DCP model, we expect that individuals with lower levels of emotional stability will experience higher levels of strain as the extent of remote work increases, regardless of their level of job autonomy. This is because they are either unaware of the existence or utility of this resource, or they perceive it as an additional threat, burden of responsibility, or need for accountability (Barsness et al., 2005; Halbesleben et al., 2014; Hobfoll, 2001; Kammeyer-Mueller et al., 2016; Rubino et al., 2012; Ten Brummelhuis & Bakker, 2012). In contrast, individuals with high levels of emotional stability can capitalize on the benefits of job autonomy as a resource, which may become more important as the extent of remote work increases (Kammeyer-Mueller et al., 2016). Consistent with the D-C and DCP models, lower levels of autonomy are likely to hamper one's ability to manage the challenges or capitalize on the benefits of remote work, particularly for those high in emotional stability. Formally, we posit:

Hypothesis 2. Emotional stability and autonomy jointly moderate the association between the extent of remote work and strain, such that a negative association only emerges among high–emotional stability individuals with high autonomy.

Three forms of need satisfaction as mediators

We have proposed a direct effect of the three-way interaction on strain, but we also seek to understand the role of three intermediary processes proposed by SDT—the degree to which needs are met for autonomy, relatedness, and competence. SDT posits that all people have these innate needs, and that fulfilment of them results in well-being and motivation (Deci & Ryan, 2000; Van den Broeck, Ferris, Chang, & Rosen, 2016). As the extent of remote work increases, we posit that these needs are harder to satisfy, particularly for employees lower in emotional stability, in turn resulting in higher levels of strain.

First, we have argued that the resource of autonomy is highly relevant and important when employees work remotely more often, but it is not inherent in remote work itself (Gajendran & Harrison, 2007; Gajendran et al., 2015). Likewise, the innate need for autonomy is not necessarily satisfied when employees work remotely. This need is directly relevant to our proposed model because perceptions of the availability of autonomy as a resource may be associated with perceptions of one's need for autonomy being satisfied, and this may also depend on one's level of emotional stability. As posited by the DCP model, employees with lower emotional stability may have a lower need and/or desire for autonomy, which will likely influence the way they experience work in general and in particular remote work. This direct link between the autonomy resource and autonomy need satisfaction is rarely empirically assessed, with scholars typically assuming it exists (e.g., Hunton, 2005). Thus, we directly test the link as we examine joint effects of the extent of remote work and emotional stability.

Second, relatedness need satisfaction is relevant to this context because isolation is a common concern when employees work remotely more often (Belle, Burley, & Long, 2015; Thatcher & Zhu, 2006). When they are physically away from the organizational setting, the availability of organizational referents for identity formation and social support is lower (Sardeshmukh et al., 2012). This means employees who work remotely more often may have weaker affective ties with the organization and co-workers, which can result in unfulfilled social relatedness needs and fewer resources overall (Colbert, Bono, & Purvanova, 2016). Indeed, all employees benefit from quality social relationships at work, rooted in an innate desire to feel included and part of a community (Belle et al., 2015; Colbert et al., 2016; O'Neill, Hambley, Greidanus, MacDonnell, & Kline, 2009). Although relationships do not have to be face-to-face for the fulfillment of relatedness needs, and those needs might be fulfilled by people in other life domains (e.g., spouse), the physical separation associated with remote work decreases the opportunities to forge social connections or accountability with co-workers, which has been shown to result in lower satisfaction and performance among remote workers (Golden, 2006b; Golden, Veiga, & Dino, 2008; O'Neill et al., 2009; Perry, Lorinkova, Hunter, Hubbard, & McMahon, 2016).

Finally, competence need satisfaction refers to how effective a person feels in interacting with their environment and in applying their own abilities towards achieving goals (Chiniara & Bentein, 2016). It is relevant to this context because employees who work remotely more often do not receive as much feedback from others, may not receive as much direct coaching or encouragement, have less visibility among co-workers, and cannot as easily compare their own abilities with their peers (Bailey & Kurland, 2002; Chiniara & Bentein, 2016; O'Neill et al., 2009; Sewell & Taskin, 2015). These processes, which could help employees develop perceptions of competence, may be thwarted when an employee works remotely more often. However, Gajendran and Harrison's (2007) meta-analysis found that supervisor–subordinate relations may actually improve and employees may view their own career prospects more favourably (which are directly related to perceptions of one's own competence when they work remotely more often). Therefore, it will be useful to examine perceptions of competence need satisfaction as a mediator to gain more insight into its role in remote work.

In line with the DCP model and the hypotheses stated earlier, we generally posit that extent of remote work is negatively associated with all three types of need satisfaction among all combinations of autonomy and emotional stability, except when autonomy and emotional stability are both high. Namely, we predict that employees with low levels of emotional stability are less able to fulfill these three needs overall, and this disadvantage is exacerbated when remote work is more frequent, even when autonomy is also high (Halbesleben et al., 2014; Hobfoll, 2001; Rubino et al., 2012). In contrast, high–emotional stability employees are better equipped to effectively utilize the job resource of autonomy to overcome any difficulties encountered in remote work, leading to fulfillment of these needs. Thus, we expect that high–emotional stability employees with high autonomy would exhibit a positive relationship between extent of remote work and autonomy, relatedness, and competence need satisfaction. In turn, SDT posits that need satisfaction is a direct predictor of employee well-being, leading us to predict a negative association between the satisfaction of each need and each strain outcome (Deci & Ryan, 2000; Halbesleben & Demerouti, 2005; Halbesleben et al., 2014; Hobfoll, 2001; Van den Broeck et al., 2016). We formally propose the following mediated-moderation model:

> ***Hypothesis 3.*** The effect of the three-way interaction is mediated by perceived need satisfaction in regard to (a) autonomy, (b) relatedness, and (c) competence, such that only those high in both job autonomy and emotional stability perceive higher need satisfaction in these areas as the extent of remote work increases. In turn, need satisfaction is negatively associated with strain.

Study 1 method

Participants and procedure

Study 1 was a two-wave data collection (3 months apart) using MarketTools Zoomerang, which rewards participants with "zoom points" that can be redeemed for rewards over time (Neubert, Kacmar, Carlson, Chonko, & Roberts, 2008). In total, 258 full-time working adults participated in both surveys (55% female, 55% individual contributors, with an average age of 45 years, 9 years organizational tenure, 43 h worked per week, and $44,882 salary). Sixty-two percent had an associate's degree or higher. We did not collect ethnicity.

Measures

Unless otherwise noted, all measures used a 5-point response scale, where 5 = *Strongly Agree*. Remote work, autonomy, and emotional stability were assessed at Time 1. Strain was assessed at Time 2.

Remote work

Following Wiesenfeld, Raghuran, and Garud (1999), we asked participants "On average, what proportion of the workweek do you spend working remotely (away from your organization's primary local office)?" Respondents used five response options, which we recoded to decimals between 0 and 1 for analysis purposes, comparable to the percentages used to reflect intensity of remote work in other published studies (Gajendran & Harrison, 2007): 1 (recoded to 0) = none; 2 (recoded to .25) = up to 25%; 3 (recoded to .50) = 26–50%; 4 (recoded to .75) = 51–75%; and 5 (recoded to 1.00) = 76–100%. Thirty-four percent of the sample reported working remotely to some extent (i.e., options 2 through 5). Of those, 38% worked remotely more than half of the time and 24% worked remotely more than three-quarters of the time. Because we were interested in the full spectrum of the extent of remote work (from none to full-time), we retained all respondents.

Autonomy

We used the three-item decision autonomy subscale from Karasek's (1979) Job Characteristics Questionnaire (α = .66; e.g., "My job allows me to make a lot of decisions on my own") to assess perceptions of the level of autonomy offered in the job.

Emotional stability

We used the 10-item subscale of the IPIP Big Five personality scale (Goldberg et al., 2006; Study 1: α = .91, Study 2: α = .90) to assess emotional stability. Respondents were given the following instructions: "Please indicate your level of agreement with each statement. Please respond with regard to how you are today, not how you wish to be or expect to be in the future. I. . .". Then a list of statements followed describing emotional stability (e.g., "… worry about things" (reversed)).

Strain

The Oldenburg Burnout Inventory assessed exhaustion (α = .83; e.g., "There are days that I feel already tired before I go to work") and disengagement (α = .82; e.g., "It happens more and more often that I talk about my work in a derogatory way") with eight items each (Demerouti, Bakker, Vardakou, & Kantas, 2002; Halbesleben & Demerouti, 2005). Job dissatisfaction was assessed using a three-item measure (Cammann et al., 1983; α = .87; e.g., "All in all, I am satisfied with my job" (reversed)).

Controls

We included gender (1 = male, 2 = female), total hours worked weekly (open-ended), salary (1 = $25,000 or less; 2 = $25,001–$50,000; 3 = $50,001–$75,000; 4 = $75,001–$100,000; 5 = $100,001–$125,000; 6 = $125,001–$150,000; and 7 = more than $150,000), and organizational tenure (open-ended in years) as controls because these are often related to strain outcomes (e.g., Huhtala, Tolvanen, Mauno, & Feldt, 2015).

Analysis

We first tested the impact of common method variance (CMV) following Podsakoff, MacKenzie, and Podsakoff (2012) and Williams, Cote, and Buckley (1989). Using confirmatory factor analysis, we allowed the survey items to load on their respective construct and on an uncorrelated latent variable, representing the method factor. The average variance explained by the method factor was 29%, and fit was not significantly better with the method factor. Hence, we do not believe CMV was more of a concern than in other published studies (Williams et al., 1989). We used hierarchical ordinary least squares regression in SAS v9.4 to test our hypotheses. All predictors except remote work, which had a meaningful zero value, were centred around the grand mean prior to creating interaction terms.

Study 1 results

See Table 1 for intercorrelations and descriptive statistics. Autonomy and emotional stability were negatively correlated with all three forms of strain, whereas remote work was only significantly correlated with disengagement ($r = -.15$, $p < .05$). We first tested Hypothesis 1 (remote work × autonomy predicts strain; see Table 2, Hypothesis 1). Although the coefficient was negative for all three outcomes, as predicted, this two-way interaction failed to reach significance for any strain outcome.

Next, we tested Hypothesis 2 (remote work × autonomy × emotional stability predicts strain; see Table 2, Hypothesis 2). The three-way interaction was significantly associated with both disengagement and dissatisfaction but not exhaustion. To understand the overall pattern of relationships, however, we plotted the interaction for all three strain outcomes using ± 1 *SD* of the mean of each moderator as cut-offs (See Figure 1 for the plots and Table 3 for slope difference tests and simple slope estimates; Dawson & Richter, 2006; Stone & Hollenbeck, 1989). The high autonomy, high emotional stability (Condition "1") was significantly different from Condition "2" (high autonomy, low emotional stability) for disengagement and dissatisfaction. For all three forms of strain, the overall level of strain for Condition "1" was lower than all other conditions. Condition "1" also had the only significant negative slope of all the conditions for disengagement. Similarly, Condition "2" had the only significant positive slope of all the conditions for dissatisfaction. For all three forms of strain, Condition "4" (low–low) exhibited higher overall levels of strain than any other condition. This pattern of results generally supported Hypothesis 2 for disengagement and dissatisfaction. Thus, the results support the notion that individuals reporting high autonomy and low emotional stability may suffer more strain when they work remotely more often, in contrast to those reporting high autonomy and high emotional stability.[2]

Study 2 method

Study 2 sought to provide further clarity on these relationships and to consider need satisfaction as a mediator. This sample included 145 full-time professional employees from three southern U.S. organizations (23% response rate; an engineering firm, a technology firm, and a local government agency representing human resource personnel, knowledge workers such as engineers and technology architects, administrative support personnel, and line managers), and it used an online survey administered at one time point. All participants had the flexibility to choose to work remotely as needed, but had one dedicated, primary office space. These organizations did not have satellite office locations, so that was not an option for working remotely. The point of contact at each organization sent an email invitation to all employees, emphasizing that participation was completely voluntary and confidential. The final sample was 41% female, 39% minority, and 60% individual contributors, with an average age of 44.3 years, 7.4 years organizational tenure, 41.9 h worked weekly, and $66,918 salary. Forty-one percent had an associate's degree or higher.

Measures

Remote work

To add more precision to our assessment of remote work over Study 1 and following common practice in assessing intensity of remote work (Gajendran & Harrison, 2007), we asked respondents two open-ended questions and then calculated percentage of time spent working remotely ("On average, how many hours do you work remotely each week (i.e., away from your organization's primary office)?" and "On average, how many total hours do you work each week?"). Overall, 53% of participants worked remotely to some extent (more than 0 h; range = 2.2–100% of the time). Of those, 25% worked remotely at least half of the time, and 8% worked remotely at least three-quarters of the time.

Autonomy

We used the 10-item Factual Autonomy Scale for a more concrete assessment of the level of the autonomy resource

Table 1. Intercorrelations and descriptive statistics: Study 1.

Variable	1	2	3	4	5	6	7	8	9	10
1. Exhaustion	(.83)									
2. Disengagement	.57***	(.82)								
3. Dissatisfaction	.54***	.66***	(.78)							
4. Remote work	−.09	−.15**	−.04	–						
5. Autonomy	−.35***	−.46***	−.41***	.12**	(.65)					
6. Emotional stability	−.58***	−.37***	−.36***	−.04	.24***	(.91)				
7. Tenure	.02	.03	−.06	−.03	−.08	.03	–			
8. Gender	.03	.02	−.06	−.07	−.01	−.09	.06	–		
9. Salary	−.20***	−.24***	−.17***	.06	.24***	.11*	−.04	−.05	–	
10. Total hours	.01	−.16***	.02	.07	.17***	.01	−.02	−.11*	.24***	–
Mean	2.73	2.79	2.18	0.19	3.56	3.31	1.80	1.56	44,882	42.61
SD	.69	.68	.94	.32	.85	.78	15.56	.50	29,864	7.69

All Likert-based items used a 5-point response scale. Alpha coefficients are presented on the diagonal. *$p < .10$; **$p < .05$; ***$p < .01$.

Table 2. Two-way and three-way interactions predicting strain outcomes (Hypotheses 1 and 2): Study 1.

Outcome	Exhaustion		Disengagement		Dissatisfaction	
	Hypothesis 1	Hypothesis 2	Hypothesis 1	Hypothesis 2	Hypothesis 1	Hypothesis 2
Predictor	*b* (*SE*)	*b* (*SE*)	*b* (*SE*)	*b* (*SE*)	*b* (*SE*)	*b* (*SE*)
Intercept	2.60*** (.25)	2.73*** (.24)	3.25*** (.26)	3.29*** (.26)	2.29*** (.38)	2.28*** (.38)
Gender	−0.03 (.07)	−0.04 (.07)	−0.02 (.08)	−0.02 (.06)	−0.15 (.11)	−0.15 (.11)
Tenure	0.00 (.00)	0.00 (.00)	−0.00 (.00)	−0.00 (.00)	−0.01 (.00)	−0.01 (.00)
Total hours worked	0.07 (.05)	0.05 (.05)	−0.06 (.05)	−0.08 (.05)	0.06 (.07)	0.06 (.07)
Salary	−0.03** (.01)	−0.02 (.01)	−0.03* (.01)	−0.02* (.01)	−0.03 (.02)	−0.03 (.02)
Remote work (A)	−0.16 (.11)	−0.15 (.12)	−0.19 (.12)	−0.09 (.13)	−0.00 (.17)	0.14 (.18)
Autonomy (B)	−0.15*** (.05)	−0.15*** (.05)	−0.29*** (.05)	−0.31*** (.05)	−0.40*** (.08)	−0.44*** (.08)
Emotional stability (C)	−0.46*** (.04)	−0.45*** (.05)	−0.26*** (.05)	−0.21*** (.06)	−0.34*** (.07)	−0.23*** (.08)
A × B	−0.03 (.14)	−0.08 (.15)	0.14 (.15)	0.12 (.16)	0.23 (.21)	0.28 (.23)
A × C		−0.00 (.16)		−0.13 (.17)		−0.38 (.25)
B × C		−0.18*** (.05)		0.01 (.06)		0.12 (.08)
A × B × C		−0.08 (.23)		−0.54** (.25)		−0.78** (.35)
R^2	.38	.41	.28	.30	.23	.25

Unstandardized regression coefficients and Adjusted R^2 reported for Study 1. For Study 1, salary and hours worked are scaled down (salary by 10,000 and hours by 10) to produce more meaningful coefficients when modelled with Likert scales. *$p < .10$; **$p < .05$; ***$p < .01$.

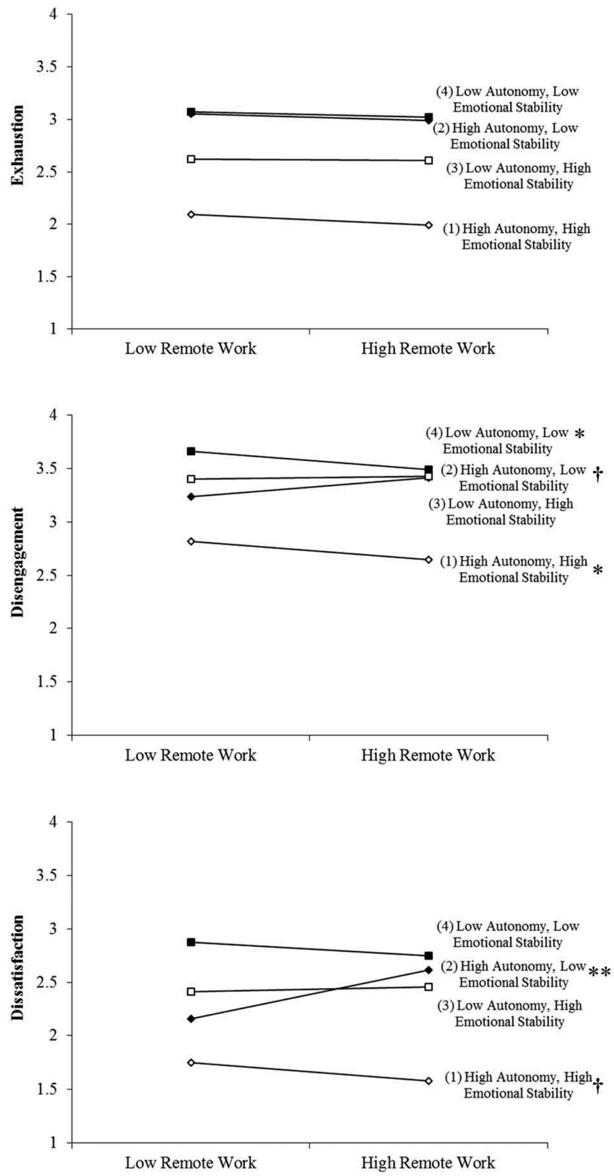

Figure 1. Hypothesis 2: three-way interaction plots predicting strain (Study 1). Symbols after the labels denote significant simple slopes: **$p < .01$; *$p < .05$; †$p < .10$.

Table 3. Slope difference tests and simple slopes for three-way interactions: Study 1.

Slope difference tests			
Outcome	Exhaustion	Disengagement	Dissatisfaction
Pair of slopes	*t*	*t*	*t*
1 and 2	−0.29	−2.45**	−3.11***
1 and 3	−0.58	−1.05	−0.88
1 and 4	−0.46	0.00	−0.27
2 and 3	−0.33	0.88	1.70*
2 and 4	−0.09	2.40**	2.71***
3 and 4	0.24	1.11	0.67

Simple slopes			
Outcome	Exhaustion	Disengagement	Dissatisfaction
Condition	Slope (*t*)	Slope (*t*)	Slope (*t*)
1	−0.31 (−1.14)	−0.64 (−2.00**)	−0.74 (−1.84*)
2	−0.15 (−0.44)	0.70 (1.91*)	1.58 (3.06***)
3	0.01 (0.02)	0.20 (0.42)	0.26 (0.40)
4	−0.15 (−0.55)	−0.62 (−1.98)	−0.54 (−1.21)

SE: standard error.
Condition 1 = high autonomy, high-emotional stability; 2 = high autonomy, low-emotional stability; 3 = low autonomy, high-emotional stability; 4 = low autonomy, low-emotional stability. ***$p < .01$; **$p < .05$; *$p < .10$.

experienced in the job compared to Study 1 (Spector & Fox, 2003; α = .73; 5 = every day; e.g., "In your present job, how often do you have to ask permission to take a rest break?").

Need satisfaction

We used the SDT needs satisfaction measure from Deci et al. (2001; 7 = very true). Namely, we used the seven-item autonomy sub-scale (α = .75; e.g., "I feel free to express my ideas and opinions on the job"), eight-item relatedness sub-scale (α = .82; e.g., "People at work care about me"), and six-item competence sub-scale (α = .67; e.g., "People at work tell me I am good at what I do").

Strain

We used the same measures as in Study 1 to measure exhaustion (eight items; α = .77), disengagement (eight items; α = .79), and dissatisfaction (three items; α = .88), with 7-point response scales (7 = "every day").

Controls

We included gender (1 = male, 2 = female), total hours worked weekly (open-ended), salary (open-ended), and permanency of work arrangement as controls. We expected permanency might affect one's sense of belonging in the organization, particularly in the context of working remotely, which lowers one's visibility ("I am confident that my current work arrangement will stay about the same for at least the next 12 months"; 5 = strongly agree; Allen et al., 2015).

Analyses

Only 7% of the variance was explained by the method factor in Study 2, reducing any concern about CMV biasing the results (Podsakoff et al., 2012; Williams et al., 1989). Because the employees were clustered in 51 distinct workgroups (delineated by supervisors), we also calculated intra-class correlations (ICC1; Bliese, 2000). ICC(1)'s were large enough to warrant concern about group effects for all outcomes and mediators (exhaustion: ICC(1) = .09; disengagement: ICC(1) = .23; dissatisfaction: ICC(1) = .13; autonomy need satisfaction: ICC(1) = .11; relatedness need satisfaction: ICC(1) = .08; competence need satisfaction: ICC(1) = .17). Thus, we proceeded using multilevel modelling with random intercepts. We used Proc Mixed in SAS v9.4 to test Hypotheses 1 and 2, and multilevel SEM (ML-SEM) in Mplus to test the path model proposed in Hypothesis 3. All predictors except remote work, which had a meaningful zero value, were centred around the grand mean prior to creating interaction terms.

Study 2 results

Table 4 presents intercorrelations and descriptive statistics for Study 2. Emotional stability was negatively correlated with all three forms of strain and all three forms of need satisfaction. Autonomy was negatively correlated with exhaustion and disengagement, as well as all three forms of need satisfaction, and the extent of remote work was not significantly correlated with any of the strain outcomes or forms of need satisfaction.

We first tested Hypothesis 1 (remote work × autonomy predicts strain; see Table 5, Hypothesis 1). Although the coefficient for the two-way interaction was negative for all outcomes, as predicted, it only reached significance for exhaustion ($b = -.22$, $p < .05$; pseudo $R^2 = .39$). The plot of this interaction in Figure 2 reveals a positive remote work-exhaustion slope among employees reporting low autonomy (0.82; $t = 2.12$, $p < .05$) and a seemingly negative but non-significant relationship among those reporting high autonomy (slope = – 1.20; $t = -1.68$, $p = .10$). This provides some support for Hypothesis 1 for exhaustion, but failed to support the hypothesis for the other forms of strain.

Next, we tested Hypothesis 2 (remote work × autonomy × emotional stability predicts strain; see Table 5, Hypothesis 2). The three-way interaction was significant in predicting disengagement (pseudo R^2 = .27) and dissatisfaction (pseudo R^2 = .20) only. As in Study 1, we proceeded to plot the interaction on all three strain outcomes to understand the overall pattern of results (see Figure 3 for plots and see Table 6 for slope difference tests and simple slope estimates).

Table 4. Intercorrelations and descriptive statistics: Study 2.

Variable	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Exhaustion	(.78)	.66***	.34***	−.37***	−.46***	−.39***	−.06	−.10	−.55***	.13	−.26**	.14	.13
2. Disengagement	.64***	(.79)	.50***	−.43***	−.45***	−.61***	−.13	−.18	−.48***	.09	−.32**	.15	.01
3. Dissatisfaction	.48***	.62***	(.88)	−.53***	−.28**	−.48**	−.16	.04	−.14	.00	−.22*	−.06	.04
4. Autonomy need satisfaction	−.37***	−.48***	−.58***	(.75)	.35***	.54***	.09	−.03	.24**	−.12	.31**	−.05	.03
5. Relatedness need satisfaction	−.32***	−.41***	−.33***	.46***	(.82)	.35***	.26**	.01	.36***	−.06	.16	−.22*	−.03
6. Competence need satisfaction	−.41**	−.66**	−.58***	.60***	.53***	(.67)	.14	.11	.35***	.16	.26**	−.03	.03
7. Remote work	.08	.04	.00	−.02	.01	−.02	–	−.22*	.13	−.14	.00	−.36***	−.06
8. Autonomy	−.19**	−.19**	−.14	.21**	.17**	.19**	.01	(.75)	−.02	.17	−.05	−.21*	.19
9. Emotional stability	−.50***	−.43***	−.23***	.29***	.31***	.40***	.04	.05	(.86)	−.13	.27**	−.03	−.11
10. Total hours	.08	−.004	−.03	−.00	−.01	.17**	−.01	.21**	−.07	–	−.08	−.03	.21*
11. Permanency	−.23***	−.21***	−.15*	.16*	.11	.13	−.12	−.02	.17*	−.08	–	.12	.06
12. Gender	.13	.20**	.03	.13	−.04	.02	−.04	−.10	.06	.03	.00	–	−.20*
13. Salary	.11	−.11	−.15	.11	.05	.11	−.15	.27***	−.07	.23**	−.06	−.34***	–
Mean	2.60	2.72	1.73	4.94	5.18	5.75	0.17	3.78	3.94	41.91	3.37	1.41	66,918
SD	0.99	1.06	0.79	1.02	0.98	0.90	0.26	0.74	0.68	7.84	1.17	0.49	41,536

All Likert-based items used a 5-point response scale except exhaustion, disengagement, and need satisfaction scales, which used a 7-point response scale. Correlations above diagonal are average within-workgroup correlations; correlations below diagonal are overall correlations, ignoring groupings. Alpha coefficients are presented on the diagonal. ***$p < .01$; **$p < .05$; *$p < .10$.

Table 5. Two-way and three-way interactions predicting strain outcomes (Hypotheses 1 and 2): Study 2.

	Exhaustion		Disengagement		Dissatisfaction	
Outcome	Hypothesis 1	Hypothesis 2	Hypothesis 1	Hypothesis 2	Hypothesis 1	Hypothesis 2
Predictor	*b* (*SE*)	*b* (*SE*)	*b* (*SE*)	*b* (*SE*)	*b* (*SE*)	*b* (*SE*)
Intercept	2.00*** (.60)	2.14*** (.61)	2.65*** (.72)	2.65*** (.72)	2.65*** (.64)	2.75*** (.59)
Gender	0.20 (.17)	0.19 (.17)	0.33* (.20)	0.38* (.20)	−0.11 (.18)	−0.10 (.16)
Permanency	−0.15** (.06)	−0.15** (.07)	−0.15* (.08)	−0.17** (.08)	−0.06 (.07)	−0.07 (.06)
Total hours worked	−0.00 (.01)	−0.00 (.01)	−0.01 (.02)	−0.01 (.02)	−0.02 (.01)	−0.02 (.01)
Salary	0.05** (.02)	0.04** (.02)	0.00 (.02)	0.01 (.02)	−0.02 (.02)	−0.02 (.02)
Remote work (A)	−0.19 (.40)	0.13 (.46)	0.34 (.48)	0.66 (.54)	−0.00 (.42)	0.52 (.44)
Autonomy (B)	−0.32***(.11)	−0.32*** (.11)	−0.25* (.13)	−0.26** (.13)	−0.14 (.11)	−0.16 (.10)
Emotional stability (C)	−0.68*** (.10)	−0.82*** (.13)	−0.67*** (.12)	−0.79*** (.16)	−0.24** (.11)	−0.49*** (.13)
A × B	−1.10** (.41)	−0.95** (.43)	−0.49 (.49)	−0.31 (.51)	−0.34 (.43)	−0.019 (.42)
A × C		−0.97 (.78)		−0.92 (.92)		−1.50* (.76)
B × C		−0.02 (.16)		0.31 (.19)		0.30* (.16)
A × B × C		−0.96* (.56)		−1.39** (.66)		−2.24*** (.54)
Pseudo R^2	.39	.39	.24	.27	.03	.20

Unstandardized coefficients from random intercepts multilevel modelling reported. Salary is scaled by 10,000.
*$p < .10$; **$p < .05$; ***$p < .01$.

Replicating Study 1, Condition "1" (high autonomy, high emotional stability) exhibited the lowest overall level of strain compared to all other conditions, for all three forms of strain. This slope was significantly negative for exhaustion and dissatisfaction (in contrast to Study 1, in which this slope was significantly negative for disengagement only).

The other results also replicated Study 1: Condition "1" was significantly different from Condition "2" (high autonomy, low emotional stability) for dissatisfaction only, Condition "2" exhibited consistently higher overall levels of strain (for all three types of strain) than Condition "1", the slope for Condition "2" was significantly positive for dissatisfaction only, and Condition "4" (low–low) exhibited higher overall levels of strain compared to the other conditions. Thus, the pattern of results provides some support for Hypothesis 2 in both studies, suggesting that indeed, the combination of high autonomy and high emotional stability may be important in protecting employee well-being as they work remotely more often. In contrast, employees who are lower in emotional stability may be less equipped to leverage job autonomy, and this may result in higher levels of dissatisfaction in particular as the extent of remote work increases. Those in the other two conditions ("3" and "4") also appear to suffer overall from more strain, and at least one form of strain may increase for these combinations as the extent of remote work increases (as shown in the simple slope estimates in Table 6).

Next, we tested Hypothesis 3 (needs satisfaction as mediators) by specifying a just-identified multilevel path model in Mplus for each mediator and each strain outcome (nine different models total; Preacher, Zyphur, & Zhang, 2010). We specified direct paths for all predictors and interaction terms on both the mediator and the strain outcome at the within-level, accounting for group-level effects in each model. As shown in the bottom section of Table 7, the three-way interaction was significant in predicting both autonomy and relatedness need satisfaction ($p < .01$).

Following the same approach we used for strain, we proceeded to plot the simple slopes at ±1 *SD* of each moderator (see plots in Figure 4 and see Table 6 for slope differences and simple slope estimates). Consistent with the results for strain in Hypothesis 2, Condition "1" (high–high) exhibited the highest overall levels of each form of need satisfaction; the slope was significantly positive for autonomy and competence need satisfaction only. In contrast, and as predicted, Condition "2" (high autonomy, low emotional stability) exhibited lower overall levels for all three forms of need satisfaction; these slopes were non-significant. Also mirroring the

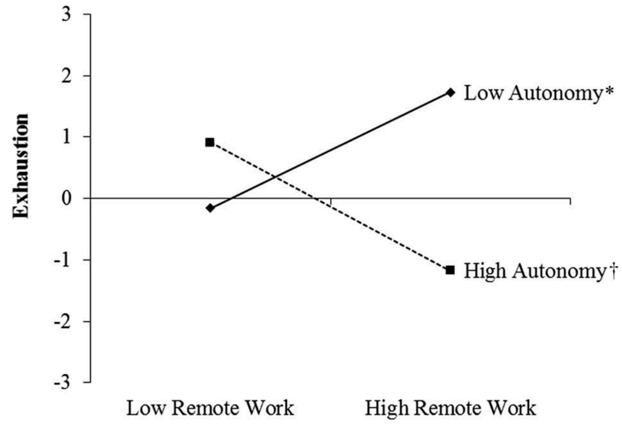

Figure 2. Hypothesis 1: two-way interaction plot predicting exhaustion (Study 2). Symbols after the labels denote significant simple slopes: *p < .05; †p = .10.

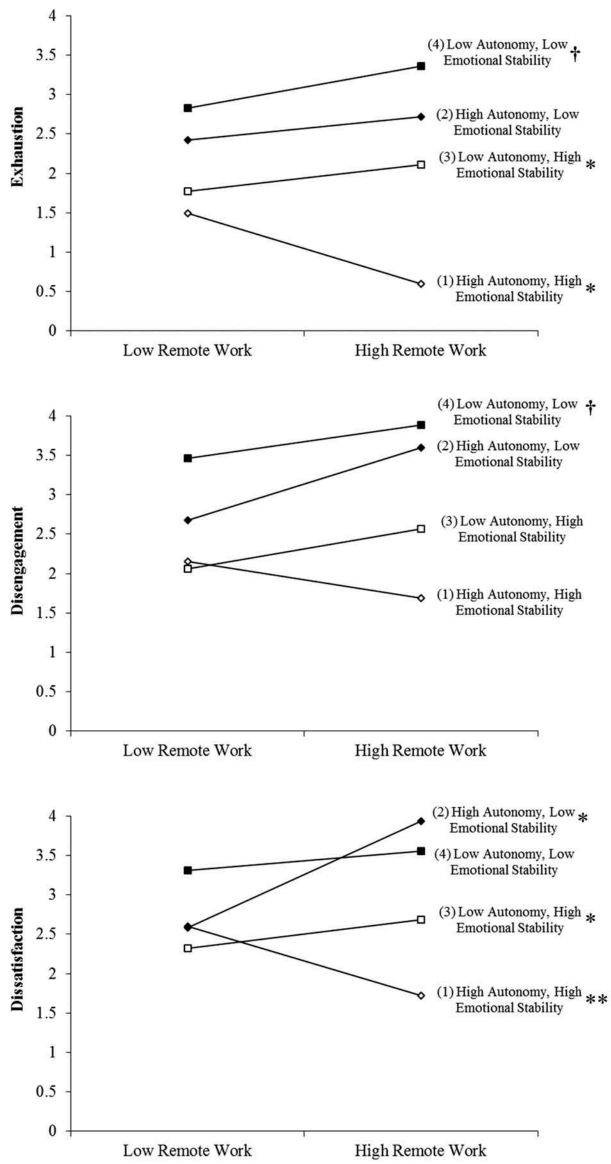

Figure 3. Hypothesis 2: three-way interaction plots predicting strain (Study 2). Symbols after the labels denote significant simple slopes: **p < .01; *p < .05; †p < .10.

Table 6. Slope difference tests and simple slopes for three-way interactions: Study 2.

			Slope difference tests			
Mediator	Exhaustion	Disengagement	Dissatisfaction	Autonomy need satisfaction	Relatedness need satisfaction	Competence need satisfaction
Pair of slopes	*t*	*t*	*t*	*t*	*t*	*t*
1 and 2	−1.48	−1.44	−2.83***	1.46	1.45	0.93
1 and 3	−3.26***	−2.16**	−3.42***	3.49***	0.78	3.35***
1 and 4	−2.64**	−1.38	−2.13**	1.09	2.45**	1.91*
2 and 3	−0.05	0.48	1.40	0.17	−1.21	0.68
2 and 4	−0.47	0.83	2.31**	−1.19	0.31	0.57
3 and 4	−0.54	0.19	0.33	−2.04**	2.98***	−0.63
			Simple slopes			
Condition	Slope (*t*)	Slope (*t*)	Slope (*t*)	Slope (*t*)	Slope (*t*)	
1	−2.76 (−2.45**)	−1.96 (−1.27)	−3.31 (−2.93***)	3.00 (2.07**)	1.90 (1.36)	2.72 (2.06**)
2	1.12 (0.59)	2.66 (1.34)	4.17 (2.41**)	−2.34 (−1.06)	−2.22 (−1.03)	−0.26 (−0.13)
3	1.06 (2.04**)	1.44 (1.53)	1.35 (2.55**)	−2.66 (−3.88***)	0.86 (1.25)	−1.66 (−2.56**)
4	1.10 (1.68*)	0.50 (1.67*)	−0.13 (−0.21)	1.00 (1.26)	−2.58 (−3.45***)	−0.40 (−0.57)

Condition 1 = high autonomy, high-emotional stability; 2 = high autonomy, low-emotional stability; 3 = low autonomy, high-emotional stability; 4 = low autonomy, low-emotional stability.
SE: standard error.
***$p < .01$; **$p < .05$; *$p < .10$.

results for strain, Condition "4" (low–low) exhibited mostly lower overall levels of all forms of need satisfaction than other conditions; it was significantly negative for relatedness need satisfaction. One unexpected finding was for autonomy need satisfaction, in which the low–low combination exhibited moderate (rather than low) levels across all levels of remote work. This suggests that employees who are lower in emotional stability may have their (lower) autonomy needs met with lower levels of autonomy. Finally, consistent with the D-C and DCP models, significant negative slopes emerged for Condition "3" (low autonomy, high emotional stability) for both autonomy and competence need satisfaction, and this condition was significantly different than Conditions "1" and/or "4" for all three forms of need satisfaction. This suggests that employees higher in emotional stability may indeed miss having higher levels of autonomy, particularly as the extent of remote work increases, and they suffer in the form of these needs going unmet.

To complete the tests of the full proposed models, we also estimated the indirect effects of the three-way interaction via the need satisfaction mediators for each strain outcome. First, as shown in the top section of Table 7, all three forms of needs satisfaction were significant at $p < .01$ in predicting all three forms of strain. For both exhaustion and disengagement, the three-way interaction exhibited an indirect effect via both autonomy and relatedness need satisfaction, but not competence. No direct effects of the three-way interaction were significant when the mediators were in the full models, suggesting full mediation via autonomy and relatedness need satisfaction in predicting both exhaustion and disengagement. For dissatisfaction, the indirect effect was significant only for autonomy need satisfaction, but a direct effect also emerged. This suggests that the joint effects of these predictors are directly associated with dissatisfaction, in part because they affect the way autonomy needs are met.[3]

Discussion

There has been a general shift towards increasing flexibility in work arrangements (e.g., telework, flexitime), which has helped organizations attract and retain top talent (Allen et al., 2012; Humphrey et al., 2007). Though it is easy to implicitly entangle the concepts of autonomy and remote work, we described how the autonomy an employee has in his/her job is distinct from the physical location in which an employee works. Our results support our propositions that these two factors independently and jointly affect employee well-being, in conjunction with the employee's level of emotional stability.

Applying the DCP model (Rubino et al., 2012) to the context of remote work arrangements, we found that the predictions of the D-C model (Karasek, 1979) only held among employees high in emotional stability. That is, employees reporting both high job autonomy and high emotional stability appear best positioned to meet their innate needs for autonomy, relatedness, and competence and in turn experience less strain overall as they work remotely more often. Significant simple slopes suggest that as remote work becomes more frequent, this may be especially true in terms of autonomy and competence need satisfaction and all three forms of strain. In contrast, employees reporting high levels of job autonomy in conjunction with lower levels of emotional stability appear less well-positioned to meet these innate needs and/or avoid strain, often regardless of the extent of remote work. It appears that more frequent remote work may be especially detrimental for these individuals' dissatisfaction, as positive slopes emerged for this outcome across both studies.

Finally, consistent with the D-C and DCP models, employees reporting low levels of both job autonomy and emotional stability may be the most susceptible to strain, regardless of how much they work remotely, as this condition exhibited the highest overall levels of strain and the lowest overall levels of relatedness and competence need satisfaction. Interestingly, this combination did exhibit a moderate level of autonomy need satisfaction; however, it is possible that their lower level of emotional stability may signal that they simply do not want or need more autonomy, and therefore, their need for this resource may be fulfilled by even low levels of it (Perry, Witt, Penney, & Atwater, 2010; Rubino et al., 2012). Employees reporting higher levels of emotional stability in conjunction

Table 7. ML-SEM results predicting needs satisfaction mediators and strain outcomes.

Outcome	Exhaustion			Disengagement			Dissatisfaction		
Mediator	Autonomy	Relatedness	Competence	Autonomy	Relatedness	Competence	Autonomy	Relatedness	Competence
Predictor	β (*SE*)	β (*SE*)	β (*SE*)	β (*SE*)	β (*SE*)	β (*SE*)	β (*SE*)	β (*SE*)	β (*SE*)
Intercept	3.46*** (.71)	3.63*** (.78)	3.20*** (.72)	4.46*** (.73)	4.51*** (.82)	5.15*** (.66)	5.76*** (.71)	5.00*** (.87)	5.78*** (.70)
Autonomy	−0.26*** (.08)			−0.40*** (.08)			−0.49*** (.08)		
Relatedness		−0.22*** (.08)			−0.31*** (.08)			−0.24*** (.09)	
Competence			−0.21*** (.08)			−0.58*** (.07)			−0.51*** (.08)
Gender	0.10 (.08)	0.04 (.08)	0.09 (.08)	0.19** (.08)	0.10 (.09)	0.17** (.07)	−0.05 (.08)	−0.11 (.09)	−0.06 (.08)
Permanency	−0.12* (.07)	−0.16** (.07)	−0.15** (.07)	−0.10 (.08)	−0.16** (.08)	−0.11* (.07)	0.00 (.08)	−0.08 (.09)	−0.04 (.08)
Total hours worked	−0.02 (.07)	−0.03 (.07)	0.03 (.08)	−0.08 (.08)	−0.09 (.08)	0.05 (.07)	−0.14* (.08)	−0.15* (.09)	−0.03 (.08)
Salary	0.21*** (.08)	0.20*** (.08)	0.20*** (.08)	0.05 (.08)	0.03 (.08)	0.03 (.07)	−0.06 (.08)	−0.09 (.09)	−0.09 (.08)
Remote work (A)	0.01 (.10)	0.00 (.10)	0.03 (.10)	0.12 (.10)	0.10 (.11)	0.14 (.09)	0.12 (.10)	0.12 (.12)	0.14 (.10)
Autonomy (B)	−0.20*** (.07)	−0.22*** (.07)	−0.21*** (.08)	−0.12 (.08)	−0.15* (.08)	−0.10 (.07)	−0.08 (.08)	−0.12 (.09)	−0.07 (.08)
Emotional stability (C)	−0.52*** (.09)	−0.50*** (.10)	−0.52*** (.10)	−0.41*** (.10)	−0.40*** (.10)	−0.29*** (.09)	−0.27*** (.10)	−0.32*** (.12)	−0.21** (.10)
A × B	−0.18** (.09)	−0.20** (.09)	−0.16* (.09)	−0.02 (.09)	−0.04 (.09)	0.07 (.08)	0.03 (.09)	−0.01 (.10)	0.10 (.10)
A × C	−0.19 (.16)	−0.12 (.16)	−0.19 (.16)	−0.15 (.16)	−0.06 (.17)	−0.12 (.15)	−0.36** (.17)	−0.30 (.19)	−0.34** (.17)
B × C	−0.10 (.09)	0.00 (.09)	−0.04 (.09)	0.05 (.10)	0.21** (.10)	0.11 (.08)	0.06 (.10)	0.24** (.11)	0.16* (.10)
A × B × C	−0.13 (.17)	−0.18 (.16)	−0.22 (.16)	−0.14 (.17)	−0.24 (.18)	−0.20 (.15)	−0.53*** (.18)	−0.72*** (.19)	−0.67*** (.17)
R^2	0.54	0.53	0.52	0.49	0.45	0.60	0.46	0.34	0.47
Total indirect effect of A × B × C via mediator	−0.17** (.07)	−.12**(.06)	−0.08 (.05)	−0.27*** (.09)	−0.16** (.07)	−0.20* (.11)	−0.32*** (.10)	−0.13* (.07)	−0.18* (.10)
	Autonomy	Relatedness	Competence						
Mediator as outcome[a]	β (*SE*)	β (*SE*)	β (*SE*)						
Intercept	4.58*** (.77)	6.23*** (.79)	4.40*** (.80)						
Gender	0.03 (.09)	−0.23*** (.09)	−0.01 (.09)						
Permanency	0.22*** (.08)	0.11 (.09)	0.14* (.08)						
Total hours worked	0.00 (.09)	−0.03 (.09)	0.22*** (.08)						
Salary	0.06 (.09)	0.00 (.10)	0.01 (.09)						
Remote work (A)	−0.06 (.12)	−0.13 (.12)	−0.02 (.11)						
Autonomy (B)	0.14* (.09)	0.10 (.09)	0.15* (.09)						
Emotional stability (C)	0.37*** (.10)	0.52*** (.11)	0.47*** (.10)						
A × B	0.12 (.10)	0.08 (.11)	0.24** (.10)						
A × C	0.10 (.19)	0.42** (.19)	0.13 (.18)						
B × C	−0.34*** (.11)	0.08 (.11)	−0.12 (.11)						
A × B × C	0.66*** (.18)	0.53*** (.19)	0.35 (.18)						
R^2	0.34	0.27	0.36						

[a]Same results for each mediator across strain outcomes.
SE: standard error.
***$p < .01$,**$p < .05$;*$p < .10$.

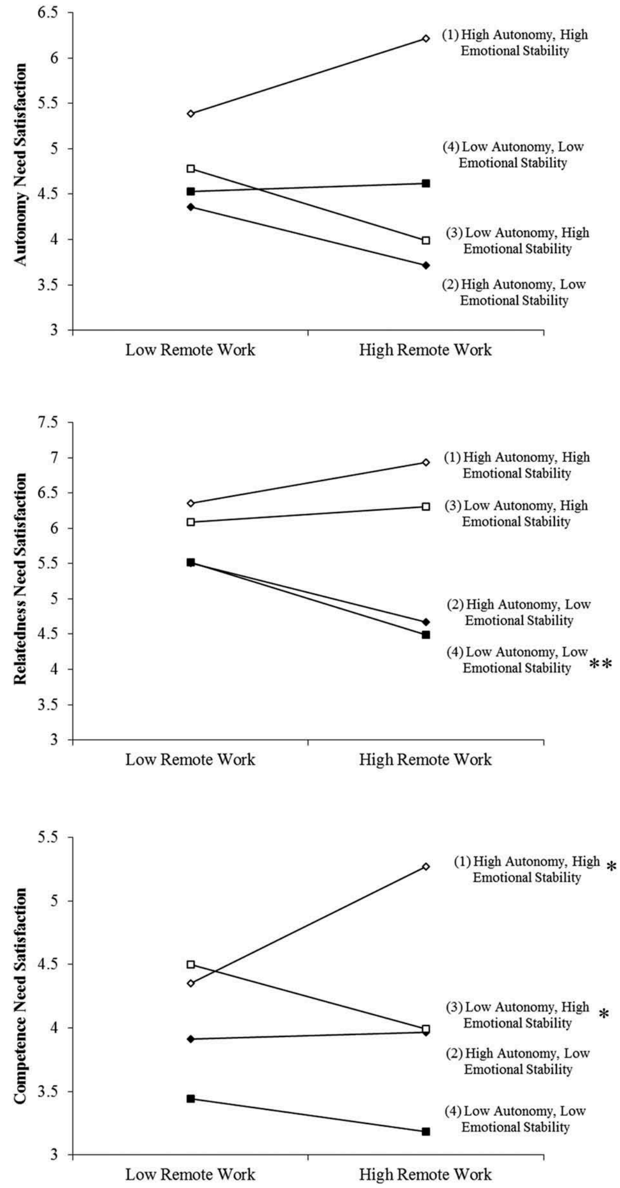

Figure 4. Hypothesis 3: three-way interaction plots predicting mediators (Study 2). Symbols after the labels denote significant simple slopes: $^{**}p < .01$; $^{*}p < .05$.

with lower levels of autonomy appeared to suffer as well, as they exhibited increases in exhaustion and dissatisfaction and decreases in autonomy and competence need satisfaction. Some nuances emerged in levels and significant slopes across samples, but the overall pattern of results was largely consistent.

Our results for the two-way interaction (remote work × autonomy) were not consistently significant, but in Study 2, a positive remote work–exhaustion slope did emerge among employees reporting low levels of autonomy. This provides some support for our propositions, but the inconsistent results reinforce the notion that emotional stability is also a critical consideration; autonomy alone may not reveal an accurate picture.

Implications for research

These results contribute to the remote work, work design, personality, and stress literatures in several ways. First, we extend research on the long-standing D-C model, and the recently expanded DCP model, to an increasingly relevant context—remote work. Our results suggest that autonomy may help employees avoid strain as they work remotely more often, particularly if emotional stability is also higher. We know remote work has both advantages and disadvantages, but these critical resources together appear to help employees manage a remote work arrangement better. Emotional stability may determine whether resources like autonomy are effectively recognized and utilized in protecting against any detrimental effects of remote work (Halbesleben et al., 2014; Meier et al., 2008; Ten Brummelhuis & Bakker, 2012). Our mostly consistent results across two studies testing the effect of the three-way interaction on strain constitute a contribution to literature on the importance of personality characteristics in stressor–strain models (Parker & Sprigg, 1999), on key considerations in flexible work design, and on the DCP model (Rubino et al., 2012).

Our results also represent a contribution to the SDT literature by explicitly testing need satisfaction for autonomy, relatedness, and competence as mediators. The findings that autonomy and relatedness need satisfaction act as mediators for exhaustion and disengagement provide evidence of these theoretical mechanisms, supported by SDT. These needs can likely be addressed through careful work design, even when employees work remotely. Furthermore, our finding that the effects on dissatisfaction were only partially mediated by autonomy need satisfaction and no other mediator suggest that the combination of remote work, autonomy, and emotional stability have important consequences for dissatisfaction directly as well as in terms of meeting needs for autonomy. This builds on recent research by empirically supporting the importance of all three needs specified by SDT in fostering positive performance-related behaviours (Chiniara & Bentein, 2016).

Thus, in applying stress–strain theories to the context of remote work, this study offers a theoretical foundation that provides critical insight into when (autonomy and emotional stability) and why (autonomy and relatedness need satisfaction) remote individuals may differentially experience three forms of strain (exhaustion, disengagement, and dissatisfaction) that comprise effective well-being and are directly linked to other important work outcomes (Mäkikangas et al., 2016).

Implications for practice

The results of these two studies reveal several clear recommendations for managers who design or oversee remote work arrangements. First, managers should consider personality when deciding which employees will work remotely at least part of the time, and how they will support those employees. In general, more emotionally stable individuals may fare better when working remotely than less emotionally stable individuals, particularly if they are also granted high levels of autonomy. But if less emotionally stable individuals must work remotely, managers should take care to provide more resources, other than autonomy, including support to help foster strong relationships with co-workers and avoid strain. Training on how to better manage demands and leverage benefits of remote work would likely be helpful. Limiting the extent of remote work and/or offering other forms of flexibility, such as flextime, may also be attractive alternatives for those who may not thrive in an intensive remote work arrangement (Allen et al., 2015; Thompson et al., 2015).

These results also highlight the importance of balancing remote work arrangements with appropriate levels of resources. Specifically, although employees may want remote work arrangements, the challenges that come with remote work may act as stressors to employees. By ensuring that these employees have appropriate levels of autonomy, managers can help mitigate some of the deleterious effects of remote work (Gajendran et al., 2015). Other resources may also help employees who work away from the office, such as proper training and equipment for remote work, separation of work and family spaces, clear procedural and performance expectations, and regular contact (virtual or face-to-face) with co-workers and managers. Therefore, managers might consider both factors when designing new types of jobs, rather than only tightening control over employees in efforts to monitor those who work remotely.

These results may also be useful in developing best practices for managing a mix of on-site and remote workers, particularly when remote work is voluntary and used differentially by employees, or even when most employees still work in the office most of the time. Any organization, regardless of the extent to which people work remotely, needs to consider well-being of their employees as they implement more flexible working practices. Our samples included a range of remote work practices which may reflect the reality of many organizations.

Limitations and future research

Although this work has a notable strength in replicating a three-way interaction across two-field samples of full-time employees, it also has a few limitations. First, all measures were self-reported. The time lag in Study 1 helps reduce this concern, however, as did the CMV analyses. Further, remote work was operationalized as percentage of time worked remotely, which does not lend itself to as much perceptual bias as affective constructs.

Second, the data were collected at one time point in Study 2 and neither study was longitudinal, which prevents us from making causal conclusions. This could also mean that people who perceive their intrinsic needs are not met are more likely to work remotely more often (perhaps "escaping" from the office) or report having lower job autonomy. Although studies have yet to explore this, O'Neill et al. (2009) found that employees who reported higher levels of need for autonomy and achievement also reported higher levels of performance when they worked remotely. Longitudinal research on our proposed models would be valuable to more fully understand the relationships between these variables. Other longitudinal research has begun to reveal useful insights; for instance, remote work increases organizational commitment and in turn, performance over time (Hunton & Norman, 2010), and employees develop spatiotemporal scaling strategies to cope with the tensions inherent in remote work (Sewell & Taskin, 2015).

Third, the overall extent of remote work in each sample may appear relatively low, since both samples had a portion of respondents who did not work remotely at all. Thus, our results could mean that people with lower emotional stability will experience more strain overall, even if they do not work remotely, and perhaps this is explained by need satisfaction. Indeed, this is consistent with research on emotional stability (Halbesleben & Buckley, 2004). A lower base rate of remote work may also explain why the two-way interaction between autonomy and remote work was non-significant in Study 1 and for two of the three outcomes in Study 2. However, the rates of remote work in our samples are consistent with recent research, Vega, Anderson, and Kaplan (2015) found that employees work remotely an average of 2.13 days per week. These are also similar to other published studies on this topic (cf. Golden, 2012; Golden & Viega, 2005; Virick, DaSilva, & Arrington, 2010). Similarly, Gajendran and Harrison (2007) meta-analysis denoted 2.5 or more days per week of remote work as "high-intensity telecommuters" and noted that 49% of their included studies used workers with 2.5 or less days per week telecommuting, suggesting that 1 or 2 days per week (or less) is an accurate reflection of the extent to which employees actually work remotely (GlobalWorkplaceAnalytics.com, 2018). Thus, due to the fact that our sample included all levels of remote work (from 0 days per week—traditional office-based workers—to 5 days per week), we cautiously suggest that our findings are indeed valuable in terms of both theory and practice.

Our results also suggest several other areas requiring further research. Different conceptualizations of the physical and/or psychological distance associated with remote work may add additional insights (cf. Bar-Anon, Liberman, & Trope, 2006). We simply asked people how often they work away from the primary office, and this may include many types of work arrangements that are very different from each other. Future research might sharpen its focus on the specific type of remote work being conducted (e.g., co-working spaces, satellite offices, customer sites, coffee shops, or at home) and how that affects relationships with co-workers, need fulfilment, and strain. For example, it is likely that working in places with other co-workers or at least friends (e.g., satellite offices or co-working spaces) might satisfy needs for relatedness and subsequently reduce strain more than isolated locations (e.g., home). In addition, assessing actual physical distance (e.g., miles) from the primary office might be useful, as well as considering media richness, frequency and type of contact, or dissimilarity among co-workers, which are all aspects and types of distance.

Other personality traits may also be important for employees who work remotely, such as extraversion, which may help individuals stay in touch with co-workers better; conscientiousness, which may help employees stay organized and productive even when working remotely; and openness to experience, which may be important for helping employees deal with a non-traditional working arrangement like remote work.[4] Future research on these traits, in concert with emotional stability, would be insightful. As the world continues to demand more flexibility from employers and employees, we are eager to continue to learn about this important managerial issue.

Notes

1. Although other personality traits may also be important in this context, we focused on emotional stability in order to directly test and extend the DCP model in conjunction with the extent of remote work (Rubino et al., 2012).
2. We also tested this three-way interaction using conscientiousness and extraversion in place of emotional stability, but neither was significant for any strain outcome. A few significant main effects emerged, as follows: for exhaustion: conscientiousness ($b = -.34$, $p < .01$); for disengagement: conscientiousness ($b = -.22$, $p < .01$) and extraversion ($b = -.21$, $p < .01$); and for dissatisfaction: conscientiousness ($b = -.29$, $p = .01$). Thus, conscientiousness and extraversion both appear to be helpful in reducing some forms of strain, but these did not alter the effects of the extent of remote work on strain in Study 1.
3. For Study 2, we also collected conscientiousness. In running the same analyses, the three-way interaction was not significant in directly predicting any form of strain (Hypothesis 2). In testing Hypothesis 3, the three-way interaction including conscientiousness *was not* significant in predicting competence or relatedness, but it *was* significant in predicting autonomy need satisfaction ($b = 2.29$, $p < .01$), and this mediator was significant in predicting each of the three strain outcomes. Only the simple slopes for Conditions "1" (high–high, positive slope) and "3" (high autonomy–low conscientiousness, negative slope) were significant at $p < .05$ and significantly different from each other ($t = 2.98$, $p < .01$). Autonomy need satisfaction was a significant mediator of the effect of this three-way interaction on all three strain outcomes (exhaustion: $\mu = -0.572$ (SE = 0.291; 95% confidence interval (CI) [−1.228, −0.103]; disengagement: $\mu = -0.939$ (*SE* = 0.403; 95% CI [−1.807, −0.229]; dissatisfaction: $\mu = -0.847$ (*SE* = 0.361; 95% CI [−1.622, −0.208]). These results mirror the results for emotional stability and provide further support for the DCP model.
4. We thank an anonymous reviewer for these suggestions.

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References

Allen, T. D., Golden, T. D., & Shockley, K. M. (2015). How effective is telecommuting? Assessing the status of our scientific findings. *Psychological Science in the Public Interest*, *16*, 40–68.

Allen, T. D., Johnson, R. C., Kiburz, K. M., & Shockley, K. M. (2012). Work-family conflict and flexible work arrangements: Deconstructing flexibility. *Personnel Psychology*, *66*, 345–376.

Ambrose, M. L., & Adler, G. S. (2000). Designing, implementing, and utilizing computerized performance monitoring: Enhancing organizational justice. *Research in Personnel and Human Resource Management*, *18*, 187–219.

Bailey, D. B., & Kurland, N. B. (2002). A review of telework research: Findings, new directions, and lessons for the study of modern work. *Journal of Organizational Behavior*, *23*, 383–400.

Bar-Anon, Y., Liberman, N., & Trope, Y. (2006). The association between psychological distance and construal level: Evidence from an implicit association test. *Journal of Experimental Psychology: General*, *135*, 609–622.

Barsness, Z. I., Diekmann, K. A., & Seidel, M.-D. L. (2005). Motivation and opportunity: The role of remote work, demographic dissimilarity, and social network centrality in impression management. *Academy of Management Journal*, *48*, 401–419.

Belle, S. M., Burley, D. L., & Long, S. D. (2015). Where do I belong? High-intensity teleworkers' experience of organizational belonging. *Human Resource Development International*, *18*, 76–96.

Bliese, P. D. (2000). Within-group agreement, non-independence, and reliability: Implications for data aggregation and analysis. In K. J. Klien & S. W. J. Kozlowski (Eds.), *Multilevel theory, research, and methods in organizations: Foundations, extensions, and new directions* (pp. 349–381). San Francisco, CA: Jossey-Bass.

Boell, S. K., Cecez-Kecmanovic, D., & Campbell, J. (2016). Telework paradoxes and practices: The importance of the nature of work. *New Technology, Work & Employment*, *31*, 114–131.

Cammann, C., Fichman, M., Jenkins, G. D., & Klesh, J. R. (1983). Assessing the attitudes and perceptions of organizational members. In S. E. Seashore, E. E. Lawler, P. H. Mirvis, & C. Cammann (Eds.), *Assessing organizational change* (pp. 71–138). New York, NY: Wiley.

Chiniara, M., & Bentein, K. (2016). Linking servant leadership to individual performance: Differentiating the mediating role of autonomy, competence and relatedness need satisfaction. *Leadership Quarterly*, *27*, 124–141.

Christian, M. S., Garza, A. S., & Slaughter, J. E. (2011). Work engagement: A quantitative review and test of its relations with task and contextual performance. *Personnel Psychology*, *64*, 89–136.

Chung-Yan, G. A. (2010). The nonlinear effects of job complexity and autonomy on job satisfaction, turnover, and psychological well-being. *Journal of Occupational Health Psychology*, *15*, 237–251.

Clear, F., & Dickson, K. (2005). Teleworking practice in small and medium-sized firms: Management style and worker autonomy. *New Technology, Work & Employment*, *20*, 218–233.

Colbert, A. E., Bono, J. E., & Purvanova, R. K. (2016). Flourishing via workplace relationships: Moving beyond instrumental support. *Academy of Management Journal*, *59*, 1199–1223.

Costa, P. T., & McCrae, R. R. (1988). Personality in adulthood: A six-year longitudinal study of self-reports and spouse ratings on the NEO Personality Inventory. *Journal of Personality and Social Psychology*, *54*, 853–863.

Crawford, E. R., LePine, J. A., & Rich, B. L. (2010). Linking job demands and resources to employee engagement and burnout: A theoretical extension and meta-analytic test. *Journal of Applied Psychology*, *95*, 834–848.

Dawson, J. F., & Richter, A. W. (2006). Probing three-way interactions in moderated multiple regression: Development and application of a slope difference test. *Journal of Applied Psychology*, *91*, 917–926.

de Jonge, J., & Kompier, M. A. J. (1997). A critical examination of the Demand-Control-Support model from a work psychological perspective. *International Journal of Stress Management*, *4*, 235–258.

Deci, E. L., & Ryan, R. M. (2000). The "what" and "why" of goal pursuits: Human needs and the self-determination of behavior. *Psychological Inquiry*, *11*, 227–268.

Deci, E. L., Ryan, R. M., Gagné, M., Leone, D. R., Usunov, J., & Kornazheva, B. P. (2001). Need satisfaction, motivation, and well-being in the work organizations of a former Eastern Bloc country. *Personality and Social Psychology Bulletin*, *27*, 930–942.

Demerouti, E., Bakker, A. B., Vardakou, I., & Kantas, A. (2002). The convergent validity of two burnout instruments: A multitrait-multimethod analysis. *European Journal of Psychological Assessment*, *18*, 29–307.

Dennis, A. R., & Wixom, B. H. (2002). Investigating the moderators of group support systems use with meta-analysis. *Journal of Management Information Systems*, *18*, 235–257.

Dimitrova, D. (2003). Controlling teleworkers: Supervision and flexibility revisited. New Technology. *Work & Employment*, *18*, 181–195.

Duxbury, L., & Halinski, M. (2014). When more is less: An examination of the relationship between hours in telework and role overload. *Work: Journal of Prevention, Assessment & Rehabilitation*, *48*, 91–103.

Gajendran, R. S., & Harrison, D. A. (2007). The good, the bad, and the unknown about telecommuting: Meta-analysis of psychological mediators and individual consequences. *Journal of Applied Psychology*, *92*, 1524–1541.

Gajendran, R. S., Harrison, D. A., & Delaney-Klinger, K. (2015). Are telecommuters remotely good citizens? Unpacking telecommuting's effects on performance via i-deals and job resources. *Personnel Psychology*, *68*, 353–393.

Giberson, T., & Miklos, S. (2013). Weighing in on telecommuting. *TIP: The Industrial-Organizational Psychologist*, *51*, 163–166.

GlobalWorkplaceAnalytics.com. (2018). *Latest telecommuting statistics*. Retrieved February 6, 2018, from http://globalworkplaceanalytics.com/telecommuting-statistics.

Goldberg, L. R., Johnson, J. A., Eber, H. W., Hogan, R., Ashton, M. C., Cloninger, C. R., & Gough, H. C. (2006). The International Personality Item Pool and the future of public-domain personality measures. *Journal of Research in Personality*, *40*, 84–96.

Golden, T., & Viega, J. F. (2005). The impact of extent of telecommuting on job satisfaction: Resolving inconsistent findings. *Journal of Management*, *31*, 301–318.

Golden, T. D. (2006a). Avoiding depletion in virtual work: Telework and the intervening impact of work exhaustion on commitment and turnover intentions. *Journal of Vocational Behavior*, *69*, 176–187.

Golden, T. D. (2006b). The role of relationships in understanding telecommuter satisfaction. *Journal of Organizational Behavior*, *27*, 319–340.

Golden, T. D. (2012). Altering the effects of work and family conflict on exhaustion: Telework during traditional and nontraditional work hours. *Journal of Business & Psychology*, *27*, 255–269.

Golden, T. D., Veiga, J. F., & Dino, R. N. (2008). The impact of professional isolation on teleworker job performance and turnover intentions: Does time spent teleworking, interacting face-to-face, or having access to communication-enhancing technology matter? *Journal of Applied Psychology*, *93*, 1412–1421.

Halbesleben, J. R. B., & Buckley, M. R. (2004). Burnout in organizational life. *Journal of Management*, *30*, 859–879.

Halbesleben, J. R. B., & Demerouti, E. (2005). The construct validity of an alternative measure of burnout: Investigating the English translation of the Oldenburg Burnout Inventory. *Work & Stress*, *19*, 208–220.

Halbesleben, J. R. B., Neveu, J.-P., Paustian-Underdahl, S. C., & Westman, M. (2014). Getting to the "COR": Understanding the role of resources in Conversation of Resources Theory. *Journal of Management*, *40*, 1334–1364.

Harmer, B. M., & Pauleen, D. J. (2012). Attitude, aptitude, ability and autonomy: The emergence of 'offroaders', a special class of nomadic worker. *Behaviour & Information Technology*, *31*, 439–451.

Hobfoll, S. E. (2001). The influence of culture, community, and the nested self in the stress process: Advancing conservation of resources theory. *Applied Psychology: An International Review*, *50*, 337–421.

Huhtala, M., Tolvanen, A., Mauno, S., & Feldt, T. (2015). The associations between ethical organizational culture, burnout, and engagement: A multilevel study. *Journal of Business and Psychology*, *30*, 399–414.

Humphrey, S. E., Nahrgang, J. D., & Morgeson, F. P. (2007). Integrating motivational, social, and contextual work design features: A meta-analytic summary and theoretical extension of the work design literature. *Journal of Applied Psychology*, *92*, 1332–1356.

Hunton, J. E. (2005). Behavioral self-regulation of telework locations: Interrupting interruptions! *Journal of Information Systems*, *19*, 111–140.

Hunton, J. E., & Norman, C. S. (2010). The impact of alternative telework arrangements on organizational commitment: Insights from a longitudinal field experiment. *Journal of Information Systems*, *24*, 67–90.

Kammeyer-Mueller, J. D., Simon, L. S., & Judge, T. A. (2016). A head start or a step behind? Understanding how dispositional and motivational resources influence emotional exhaustion. *Journal of Management*, *42*, 561–581.

Karasek, R. A. (1979). Job demands, job decision latitude, and mental strain: Implications for job redesign. *Administrative Science Quarterly*, *24*, 285–308.

Karasek, R. A., & Theorell, T. (1990). *Healthy work, stress, productivity, and the reconstruction of working life*. New York, NY: Basic Books.

Kelliher, C., & Anderson, D. (2010). Doing more with less? Flexible working practices and the intensification of work. *Human Relations*, *63*, 83–106.

Konradt, U., Hertel, G., & Schmook, R. (2003). Quality of management by objectives, task-related stressors, and non-task-related stressors as predictors of stress and job satisfaction among teleworkers. *European Journal of Work and Organizational Psychology*, *12*, 61–79.

Kühnel, J., Sonnentag, S., & Bledow, R. (2012). Resources and time pressure as day-level antecedents of work engagement. *Journal of Occupational & Organizational Psychology*, *85*, 181–198.

Lapierre, L. M., Van Steenbergen, E. F., Peeters, M. W., & Kluwer, E. S. (2016). Juggling work and family responsibilities when involuntarily working more from home: A multiwave study of financial sales professionals. *Journal of Organizational Behavior*, *37*, 804–822.

Luchman, J. N., & González-Morales, M. G. (2013). Demands, control, and support: A meta-analytic review of work characteristics interrelationships. *Journal of Occupational Health Psychology*, *18*, 37–52.

Mäkikangas, A., Kinnunen, U., Feldt, T., & Schaufeli, W. (2016). The longitudinal development of employee well-being: A systematic review. *Work & Stress*, *30*, 46–70.

Mauno, S., Mäkikangas, A., & Kinnunen, U. (2016). A longitudinal person-centred approach to the job demands-control model. *European Journal of Work and Organizational Psychology*, *25*, 914–927.

McNaughton, D., Rackensperger, T., Dorn, D., & Wilson, N. (2014). 'Home is at work and work is at home': Telework and individuals who use augmentative and alternative communication. *Work: Journal of Prevention, Assessment & Rehabilitation*, *48*, 117–126.

Meier, L., Semmer, N. K., Elfering, A., & Jacobshagen, N. (2008). The double meaning of control: Three-way interactions between internal resources, job control, and stressors at work. *Journal of Occupational Health Psychology*, *13*, 244–258.

Michel, J. S., Kotrba, L. M., Mitchelson, J. K., Clark, M. A., & Baltes, B. B. (2011). Antecedents of work-family conflict: A meta-analytic review. *Journal of Organizational Behavior*, *32*, 689–725.

Neubert, M. J., Kacmar, K. M., Carlson, D. S., Chonko, L. B., & Roberts, J. A. (2008). Regulatory focus as a mediator of the influence of initiating structure and servant leadership on employee behavior. *Journal of Applied Psychology*, *93*, 1220–1233.

O'Neill, T. A., Hambley, L. A., Greidanus, N. S., MacDonnell, R., & Kline, T. J. B. (2009). Predicting teleworker success: An exploration of personality, motivational, situational, and job characteristics. *New Technology, Work & Employment*, *24*, 144–162.

Parker, S. K., & Sprigg, C. A. (1999). Minimizing strain and maximizing learning: The role of job demands, job control, and proactive personality. *Journal of Applied Psychology*, *84*, 925–939.

Perry, S. J., Lorinkova, N. M., Hunter, E. M., Hubbard, A., & McMahon, J. T. (2016). When does virtuality really "work"? Examining the role of work-family and virtuality in social loafing. *Journal of Management*, *42*, 449–479.

Perry, S. J., Witt, L. A., Penney, L. M., & Atwater, L. (2010). The downside of goal-focused leadership: The role of personality in subordinate exhaustion. *Journal of Applied Psychology*, *95*, 1145–1153.

Peters, V., Houkes, I., De Rijk, A. E., Bohle, P. L., Engels, J. A., & Nijhuis, F. J. (2016). Which resources moderate the effects of demanding work schedules on nurses working in residential elder care? A longitudinal study. *International Journal of Nursing Studies*, *58*, 31–46.

Pisanti, R., Van Der Doef, M., Maes, S., Lombardo, C., Lazzari, D., & Violani, C. (2015). Occupational coping self-efficacy explains distress and well-being in nurses beyond psychosocial job characteristics. *Frontiers in Psychology*, *6*, 1143.

Podsakoff, P. M., MacKenzie, S. B., & Podsakoff, N. P. (2012). Sources of method bias in social science research and recommendations on how to control it. *Annual Review of Psychology*, *65*, 539–569.

Preacher, K. J., Zyphur, M. J., & Zhang, Z. (2010). A general multilevel SEM framework for assessing multilevel mediation. *Psychological Methods*, *15*, 209–2033.

Quinn, R. W., Spreitzer, G. M., & Lam, C. F. (2012). Building a sustainable model of human energy in organizations: Exploring the critical role of resources. *Academy of Management Annals*, *6*, 337–396.

Rousseau, D. M., Ho, V. T., & Greenberg, J. (2006). I-deals: Idiosyncratic terms in employment relationships. *Academy of Management Review*, *31*, 977–994.

Rousseau, D. M., Tomprou, M., & Simosi, M. (2016). Negotiating flexible and fair idiosyncratic deals (i-deals). *Organizational Dynamics*, *45*, 185–196.

Rubino, C., Perry, S. J., Milam, A. C., Spitzmüller, C., & Zapf, D. (2012). Demand–Control–Person: Integrating the demand–Control and conservation of resources models to test an expanded stressor–Strain model. *Journal of Occupational Health Psychology*, *17*, 456–472.

Sardeshmukh, S. R., Sharma, D., & Golden, T. D. (2012). Impact of telework on exhaustion and job engagement: A job demands and job resources model. *New Technology, Work & Employment*, *27*, 193–207.

Sewell, G., & Taskin, L. (2015). Out of sight, out of mind in a new world of work? Autonomy, control, and spatiotemporal scaling in telework. *Organization Studies*, *36*, 1507–1529.

Shamir, B., & Salomon, I. (1985). Work-at-home and the quality of working life. *Academy of Management Review*, *10*, 455–464.

Shumate, M., & Fulk, J. (2004). Boundaries and role conflict when work and family are colocated: A communication network and symbolic interaction approach. *Human Relations*, *57*, 55–74.

Society for Human Resource Management. (2017). *SHRM customized employee benefits prevalence benchmarking report*. Alexandria, VA. Retrieved February 16, 2018, from https://www.shrm.org/ResourcesAndTools/business-solutions/Documents/Benefits-Prevalence-Report-All-Industries-All-FTEs.pdf

Spector, P. E., & Fox, S. (2003). Reducing subjectivity in the assessment of the job environment: Development of the Factual Autonomy Scale (FAS). *Journal of Organizational Behavior*, *24*, 417–432.

Stone, E. F., & Hollenbeck, J. R. (1989). Clarifying some controversial issues surrounding statistical procedures for detecting moderator variables, Empirical evidence and related matters. *Journal of Applied Psychology*, *74*, 3–10.

Ten Brummelhuis, L. L., & Bakker, A. B. (2012). A resource perspective on the work-home interface. *American Psychologist*, *67*, 545–556.

Thatcher, S. M. B., & Zhu, X. (2006). Changing identities in a changing workplace: Identification, identity enactment, self-verification, and telecommuting. *Academy of Management Review*, *31*, 1076–1088.

Thompson, R. J., Payne, S. C., & Taylor, A. B. (2015). Applicant attraction to flexible work arrangements: Separating the influence of flextime and flexplace. *Journal of Occupational & Organizational Psychology*, *88*, 726–749.

Van den Broeck, A., De Cuyper, N., & De Witte, H., & Vansteenkiste, *M*. (2010). Not all job demands are equal: Differentiating job hindrances and job challenges in the Job Demands-Resources model. European Journal of Work and Organizational Psychology, 19, 735 – 759

Van den Broeck, A., Ferris, D. L., Chang, C.-H., & Rosen, C. C. (2016). A review of Self-Determination Theory's basic psychological needs at work. *Journal of Management*, *42*, 1195–1229.

Vega, R. P., Anderson, A. J., & Kaplan, S. A. (2015). A within-person examination of the effects of telework. *Journal of Business and Psychology*, *30*, 313–323.

Virick, M., DaSilva, N., & Arrington, K. (2010). Moderators of the curvilinear relation between extent of telecommuting and job and life satisfaction: The role of performance outcome orientation and worker type. *Human Relations*, *63*, 137–154.

Wiesenfeld, B., Raghuram, S., & Garud, R. (1999). Communication patterns as determinants of organizational identification in a virtual organization. *Organization Science*, *10*, 777–790.

Williams, L. J., Cote, J. A., & Buckley, M. R. (1989). Lack of method variance in self-reported affect and perceptions at work: Reality or artifact. *Journal of Applied Psychology*, *74*, 462–468.

Wright, T. A., & Cropanzano, R. (1998). Emotional exhaustion as a predictor of job performance and voluntary turnover. *Journal of Applied Psychology*, *83*, 486–493.